MUHAMMED SALIH YILMAZ

O-notation

$O(g(n)) = \{ f(n)$. there exist positive constants c and n_0 such that;
$0 \leq f(n) \leq c.g(n)$ for all $n \geq n_0$ $\}$

a) $(2^n+n^3) \in O(4^n) \Rightarrow O(4^n) = \{ 2^n+n^3 ;\ 0 \leq (2^n+n^3) \leq c.4^n$

we can put any(+)number in constant c, and the equation would still hold.

such as $0 \leq (2^n+n^3) \leq 1.4^n$
$\leq 3.4^n$ $\Rightarrow$ Since $2^n > n^3$
we can compare 2^n with 4^n ($\downarrow$ 2^{2n})
2^n is always smaller than 2^{2n}

a is true

---

b) $\sqrt{10n^2+7n+3} \in \Omega(n)$

Ω - notation

$\Omega(g(n)) = \{ f(n)$: there exists positive constants c and n_0 such that;
$0 \leq c.g(n) \leq f(n)$ for all $n \geq n_0$ $\}$

$\sqrt{10n^2+7n+3} \in \Omega(n) \Rightarrow \Omega(n) = \{ \sqrt{10n^2+7n+3} ;\ 0 \leq c.n \leq \sqrt{10n^2+7n+3}$

we can simplify the given notation to $\sqrt{10n^2} \Rightarrow (10.n)^{2/2} = n\sqrt{10} \rightarrow n \rightarrow$ since $10n^2$ is the largest element. Thus;

$\Omega(n) = \{ n\sqrt{10} ;\ 0 \leq c.n \leq n\sqrt{10} \}$
for all $n \geq n_0$

if we choose $n_0 = \sqrt{10}$; then for all the numbers larger than $\sqrt{10}$; the inequality will not hold.
Such as $c = \sqrt{12} \rightarrow n.\sqrt{12} \leq n.\sqrt{10}$ X

Thus, b is FALSE

This statement should be $\Theta(n)$ not $\Omega(n)$

2

c) o-notation (Little o)

$o(g(n)) = \{ f(n) :$ for all constants $c > 0$, there exists a constant $n_0 > 0$; such that $0 \leq f(n) \leq c.g(n)$ for all $n \geq n_0 \}$.

$(n^2 + n) \in o(n^2)$

$o(n^2) = \{ n^2+n \; ; \; 0 \leq n^2+n \leq c.n^2 \}$ for all $n > n_0 \}$

if we choose $n_0 = 1$, we find that every constant number we put to c, will result in the inequality to be true. Such as $n = 2, 3, 4 \ldots$

$0 \leq n^2+n \leq 2n^2 \rightarrow$ $n^2 \leq 2n^2$ ✓

$n^2 \leq 3n^2$ ✓

...

Thus; it holds

Thus, c is FALSE ← But when $c = 1 \rightarrow$ It doesn't hold.

When we find 1 contradiction, it is game over

$n^2 + n \leq 1.n^2 \longrightarrow$ ✗ False

d) Θ-notation

$\Theta(g(n)) = \{ f(n) :$ there exists positive constants c_1, c_2 and n_0 such that; $0 \leq c_1.g(n) \leq f(n) \leq c_2.g(n)$ for all $n > n_0 \}$

$3 \log^2 n \in \Theta(\log n^2)$

$\Theta(\log n^2) = \{ 3 \log^2 n \; ; \; 0 \leq c_1. \log n^2 \leq 3 \log^2 n \leq c_2. \log n^2$

if we choose $c_1 = 1$ and $c_2 = 4$ and $n_0 = 5$

$\log n^2 \leq 3 \log^2 n \leq 4 \log n^2 \rightarrow$ we see that this does not hold

eventually $3 \log^2 n$ becomes the slowest of the bunch.

Thus; $3 \log^2 n \notin \Theta(\log n^2)$ FALSE

e) Big-O notation

$O(g(n)) = \{ f(n)$. there exists positive constants c and n_0 such that ;
$0 \leq f(n) \leq c \cdot g(n)$ for all $n \geq n_0 \}$

$(n^3+1)^6 \in O(n^3)$

↳ The biggest element is what matters and the rest of the equation is unimportant.

$(n^3+1)^6 = n^{18} + 6n^{15} + \ldots \Rightarrow$ biggest element $\underline{n^{18}}$

$O(n^3) = \{ n^{18} ; 0 \leq n^{18} \leq c \cdot n^3$ for all $n \geq n_0 \}$

no matter what constant we choose (after $n_0 = 1$) ; n^{18} will always be larger than n^3

$c = 3, 4$ $\quad 3n^{18} \leq n^3$ ✗ False

Thus ; $\underline{e \text{ is not true}}$

QUESTION 2

a) $2n \log(n+2)^2 + (n+2)^2 \log\frac{n}{2} = \underbrace{2n \cdot 2\log(n+2)}_{} + \underbrace{(n+2)^2 \cdot (\log n - 1)}_{}$ when we simplify

$\in \Theta(n \cdot \log n) + \Theta(n^2 \log n) = \underline{\Theta(n^2 \log n)}$

$g(n) = n^2 \log n$

we take the largest of the elements ; $n^2 \log n$ is larger than $n \log n$.

b) $0.001 n^4 + 3n^3 + 1$

↳ in Θ-notation, we must look for the dominant term ; which is $\boxed{0,001n^4}$ here.

$\Theta(0.001 n^4) = \Theta(c \cdot n^4) = \underline{\Theta(n^4)}$ $\quad g(n) = n^4$

CamScanner ile tarandı

QUESTION 3

a) $\log n,\ n^{\log n},\ n^{1,5}$

$$\lim_{n\to\infty} \frac{\log n}{n^{\log n}} \to \text{L'hospital} \to \lim_{n\to\infty}\left(\frac{\frac{1}{n.\cancel{\ln 2}}}{\frac{2.n^{\log_2^{n-1}}.\ln n}{\cancel{\ln 2}}}\right) = \lim_{n\to\infty} \frac{1/n}{2.n^{\log_2^{n-1}}.\ln n}$$

$$\lim_{n\to\infty} \frac{1}{2.n^{\log_2^n}.\ln n} = \frac{1}{\infty} = 0 \to$$ Sonuç $0 \Rightarrow$ üstteki hızlı
Sonuç $\infty \Rightarrow$ alttaki hızlı

$$\lim_{n\to\infty} \frac{n^{\log n}}{n^{1.5}} = \frac{1}{n^{1.5}} . \lim_{n\to\infty} (n^{\log n})$$

$$= \frac{1}{n^{1.5}} . \lim_{n\to\infty} \left(e^{\log n . \ln n}\right) = \frac{1}{n^{1.5}} . \infty = \infty$$ ↓ alttaki hızlı.

Thus;

$$\log n < n^{1,5} < n^{\log n}$$

c) $n\log n,\ \sqrt{n}$

$$\lim_{n\to\infty} \frac{n\log n}{\sqrt{n}} = \lim_{n\to\infty} \frac{n}{\sqrt{n}} . \log n = \lim_{n\to\infty} n^{1/2} . \log n = \infty \Rightarrow$$ ↓ alttaki hızlı

$$\sqrt{n} < n.\log n$$

d) $n.2^n,\ 3^n$

$$\lim_{n\to\infty} \frac{3^n}{2^n.n} \Rightarrow \lim_{n\to\infty} \frac{(1,5)^n}{n} \to \text{L'hospital} \to \lim_{n\to\infty} \frac{(1,5)^n . \ln(1,5)}{1} = \infty$$

$\infty \to$ ↓ alttaki hızlı

$$n.2^n < 3^n$$

e) $\sqrt{n+10},\ n^3$

$$\lim_{n\to\infty} \frac{\sqrt{n+10}}{n^3} \to \lim_{n\to\infty}\left[\frac{n^3\sqrt{\frac{1}{n^5}+\frac{10}{n^6}}}{n^3}\right] = \lim_{n\to\infty} \sqrt{\frac{1}{n^5}+\frac{10}{n^6}} = \sqrt{0+10.0} = 0 \to$$ ↑ üstteki hızlı

$$\sqrt{n+10} < n^3$$

b) $n!$, 2^n, n^2

$$\lim_{n\to\infty} \frac{2^n}{n^2} = \frac{\infty}{\infty} \to \text{L'hospital} \to \frac{2^n . \overset{\text{sabit}}{\ln 2}}{2.n} \to \frac{\ln 2}{2} . \lim_{n\to\infty} \frac{2^n}{n} \to \text{L'hospital}$$

$$\lim_{n\to\infty} \frac{2^n . \ln 2}{1} \to \ln 2 . \lim_{n\to\infty} 2^n = \infty \to \downarrow \text{alttaki hızlı.}$$

$$n^2 < 2^n$$

Stirling Formula

$$\ln(n!) = \ln 1 + \ln 2 + \dots + \ln n$$
$$= \sum_{j=1}^{n} \ln j \approx \int_1^n \ln j = n.\ln(n) - n + 1$$
$$\approx n.\ln(n) - n$$

$$\lim_{n\to\infty} \frac{\cancel{n}.(\ln n - 1)}{n^2} = \infty \to n^2 < n!$$

Comparison with $n!$ and 2^n

$$n! = \underbrace{\overbrace{n.(n-1).\dots\dots}^{S(1)}}_{n-4 \text{ term}} \overbrace{4.3.2.1}^{24}$$
$$2^n = \underbrace{2.2.2\dots\dots}_{\substack{n-4 \text{ term} \\ S(2)}} \underbrace{2.2.2.2}_{16}$$

When we compare $S(1)$ and $S(2)$ we see that every single term of $S(1)$ is bigger than of $S(2)$. In the last 4 digits, 24 is also bigger than 16.

Thus $S(1) > S(2)$

Thus; $\underline{n^2 < 2^n < n!}$

Finally when we compare all of the functions, we get;

$$n! > 3^n > n.2^n > 2^n > n^3 > n^2 > n^{\log n} > n^{1.5} > n.\log n > \sqrt{n} > \log n$$

QUESTION 4

a) Worst case occurs when; all the element are equal or last two elements are the only elements that are not equal.

The basic operation is the comparison in the inner loop → B[i,j] != B[j,i]

b) $T_{worst}(n) = \sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} = \sum_{i=0}^{n-2} [(n-1)-(i+1)+1] = \sum_{i=0}^{n-2} (n-1+i)$

$= \frac{n.(n-1)}{2}$

The algorithm must executed $\frac{n.(n-1)}{2}$ times at worst

c) Since we know the summation expression; we can derive the complexity.

$\frac{n.(n-1)}{2} \rightarrow \frac{n^2-n}{2} \rightarrow \Theta\left(\frac{n^2-n}{2}\right) = \Theta(n^2)$

QUESTION 5

a) Basic operation is the most important operation of the algorithm that contributes to the most running time.

The basic operation of this algorithm is in the innermost loop;

$$C[i,j] = C[i,j] + A[i,k] * B[k,j]$$

b) $\sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} 1 = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} n = \sum_{i=0}^{n-1} n^2 \Rightarrow n^3$

c) Since we know the summation relation, we can derive the complexity.

$n^3 \rightarrow \Theta(n^3)$

QUESTION 6

```
algorithm ( array[], sizeOfArray, desiredNumber):
    for i=0 to sizeOfArray do:
        for j=i to sizeOfArray do:
            if (array[i] * array[j] == desiredNumber):
                print (array[i], array[j])
```

In this algorithm, we have 2 nested loops. Generally in these circumstances, the answer would be $O(n^2)$, but we must prove it.

First for loop runs for n times. $\rightarrow O(n)$

Second for loop runs for $n/2$ times, because j starts from i, not $0 \rightarrow O(n)$

if statement includes a constant time operation $\rightarrow O(1)$

Thus; $\quad n \cdot \frac{n}{2} = \frac{n^2}{2} \rightarrow O\left(\frac{n^2}{2}\right) \Rightarrow \underline{O(n^2)}$